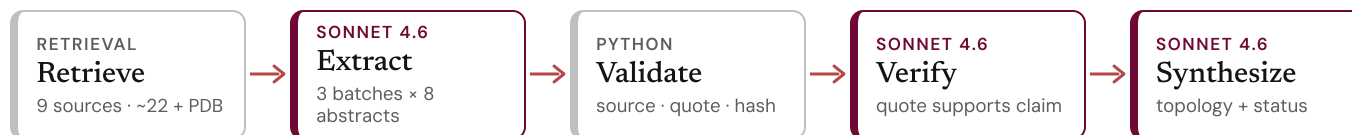


Surface proteome annotation with per-claim citation integrity.

A Claude-driven re-annotation of the ~9,000-gene human surface proteome — correcting known errors in SURFY / CSPA / UniProt / GO, adding graded evidence confidence and topology, and shipping every claim with a verbatim literature span. Public blog post, code, and results table.

Per-gene pipeline SONNET-FIRST · OPUS ON UNCERTAINTY



ROUTE BY **Sonnet confidence** · **DB agreement** (SURFY / CSPA / UniProt / GO) · **verification result** · **primary-evidence count**

ESTIMATED ROUTING DISTRIBUTION ACROSS 9K GENES

CONFIDENT · DBS AGREE
~60%
 Persist · Sonnet resolved

DB DISAGREE · MED/LOW CONFIDENCE
~30%
 Opus light · re-synthesize same evidence → persist

CONTRADICTION · DARK GENE · FAILED VERIFICATION
~10%
 Opus heavy · +2–3 OA full papers · re-extract → persist

Per-gene token budget SONNET 4.6

PHASE	INPUT	OUTPUT
Evidence extraction (3 × 8 abstracts)	20.1k	6.3k
Full-text methods (~1.5 papers)	16.5k	3.0k
Claim verification	18.5k	1.5k
Gene synthesis	10.5k	1.5k
Sonnet per-gene total	~66k	~12k

Opus cascade OPUS 4.7

PATH	SHARE	INPUT	OUTPUT
Light — re-synthesize same evidence	30%	~15k	~4k
Heavy — + 2–3 OA full papers	10%	~35k	~7k
Averaged across all 9k genes	40%	~8k	~2k

- Cascade triggered by DB disagreement, contradictory evidence, or failed claim verification — 60% of genes resolve on Sonnet alone.

ESTIMATED CREDIT ENVELOPE (BASE SCENARIO, 9K GENES)

~670M input · ~125M output

At current list rates: **≈ \$6,400 end-to-end**. Tokens are our primary planning unit; we'll use the Batch API where possible to reduce cost further.

Scenario envelope 9,000 GENES

SCENARIO	OPUS MIX (LIGHT / HEAVY)	TOTAL INPUT TOKENS	TOTAL OUTPUT TOKENS	≈ COST
Tight	20% / 5%	~620M	~120M	\$5,900
Base	30% / 10%	~670M	~125M	\$6,400
Loose	40% / 15%	~740M	~135M	\$6,900

- **100-gene pilot**: ~7M in · ~1.3M out · ~\$55.